

The Creative Workshop & Studio Management

TA : Dina Amr

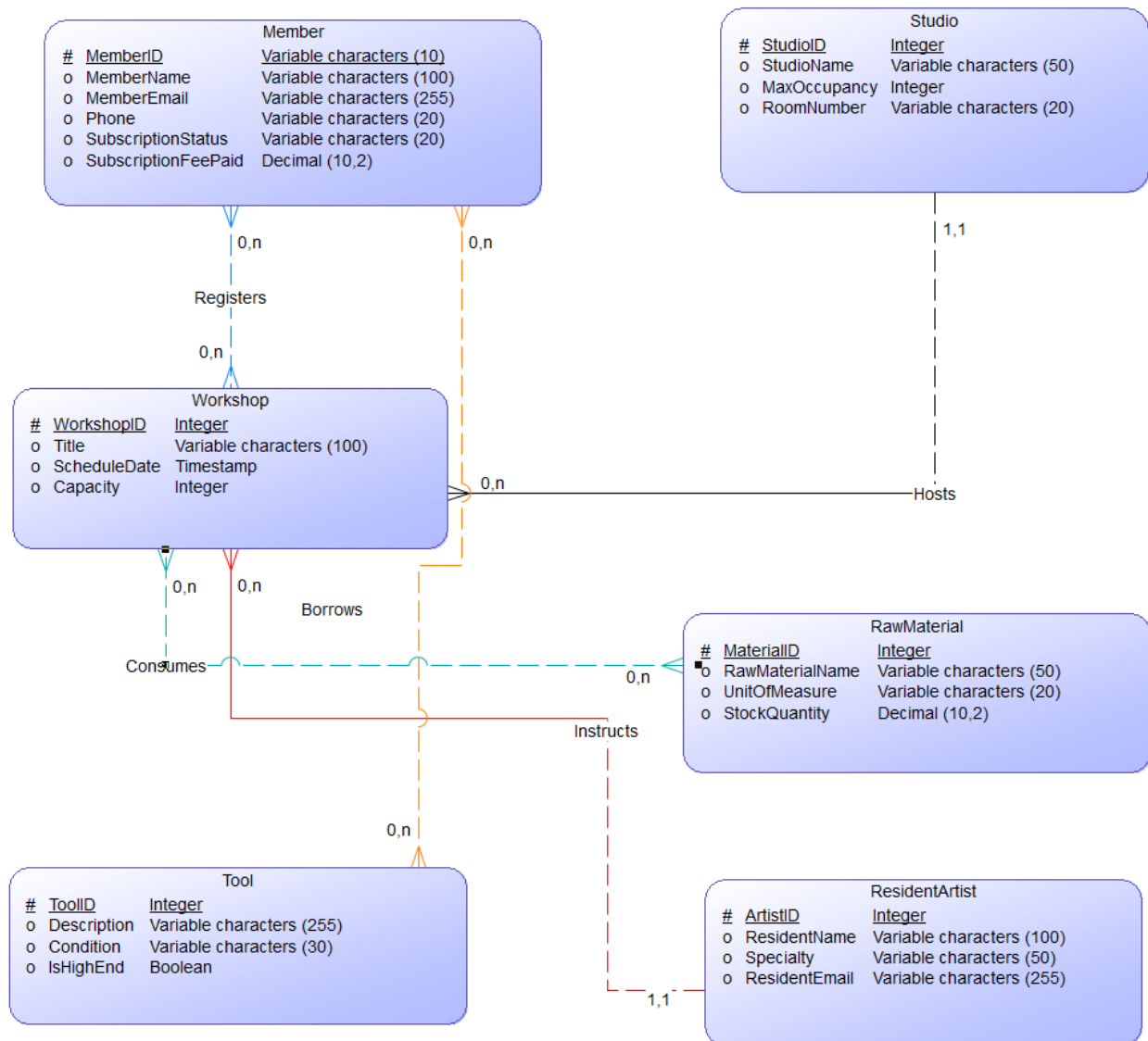
ProjectID:313

Name – ID – Lab Group :

- **Mena Sultan - 20245068 – G1**
- **Ahmed Hasanien - 20246011 – G12**
- **Esmail Khetam - 20246019 – G12**
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- **Mohamed Elnaggar - 20247006 – G12**
- **Sayed Ahmed – 20257036 - All (G12)**

Program : Special Zayed

1.Full ERD (Conceptual Model):



2.All Entities :

1.Member:

Entity Properties - Member (MEMBER)

General Attributes Identifiers Subtypes Notes Rules

1 MemberID MEMBERI Variable 10 ☒ ☒ ☒ <None>

2 MemberNam MEMBER Variable 100 ☐ ☐ ☒ <None>

3 MemberEma MEMBER Variable 255 ☐ ☐ ☒ <None>

4 Phone PHONE Variable 20 ☐ ☐ ☒ <None>

5 Subscription SUBSCRI Variable 20 ☐ ☐ ☒ <None>

6 Subscription SUBSCRI Decimal 10 2 ☐ ☐ ☒ <None>

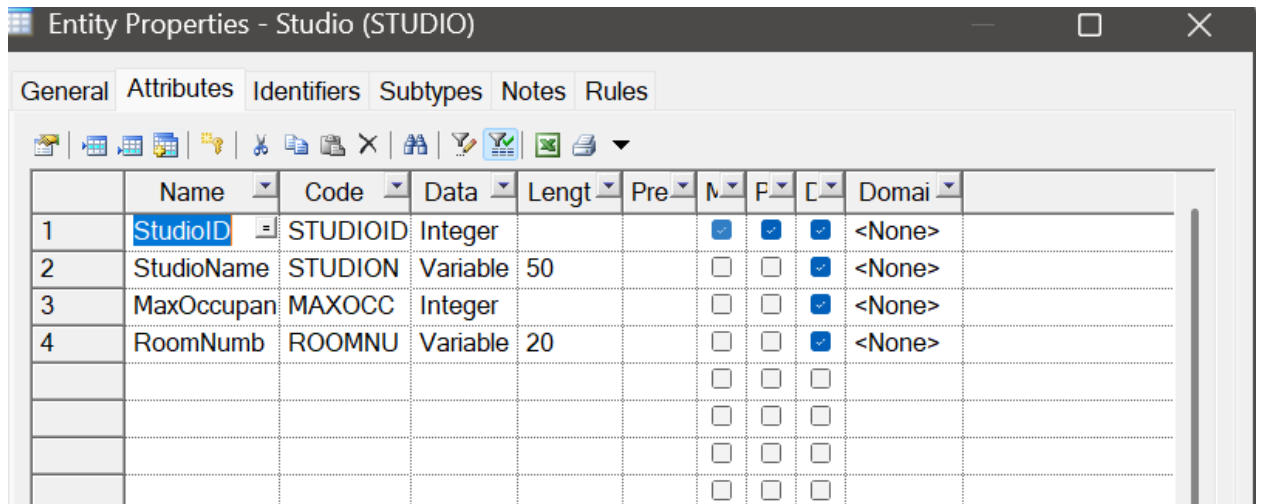
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Entity Properties - Member (MEMBER)

General Attributes Identifiers Subtypes Notes Rules

1 MemberID MEMBERID ☒

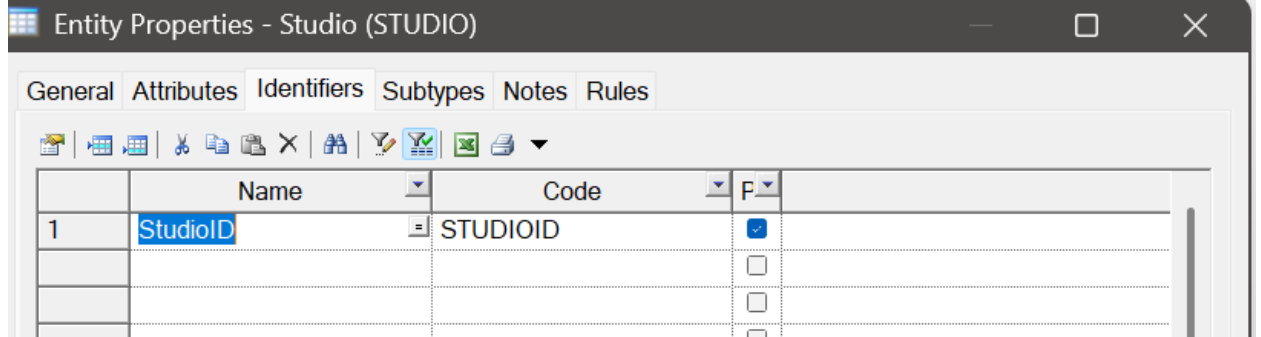
2. Studio:



Entity Properties - Studio (STUDIO)

General Attributes Identifiers Subtypes Notes Rules

	Name	Code	Data	Length	Pre	M	P	C	Domain
1	StudioID	STUDIOID	Integer			☒	☒	☒	<None>
2	StudioName	STUDION	Variable	50		☐	☐	☒	<None>
3	MaxOccupan	MAXOCC	Integer			☐	☐	☒	<None>
4	RoomNumb	ROOMNU	Variable	20		☐	☐	☒	<None>
						☐	☐	☐	
						☐	☐	☐	
						☐	☐	☐	
						☐	☐	☐	

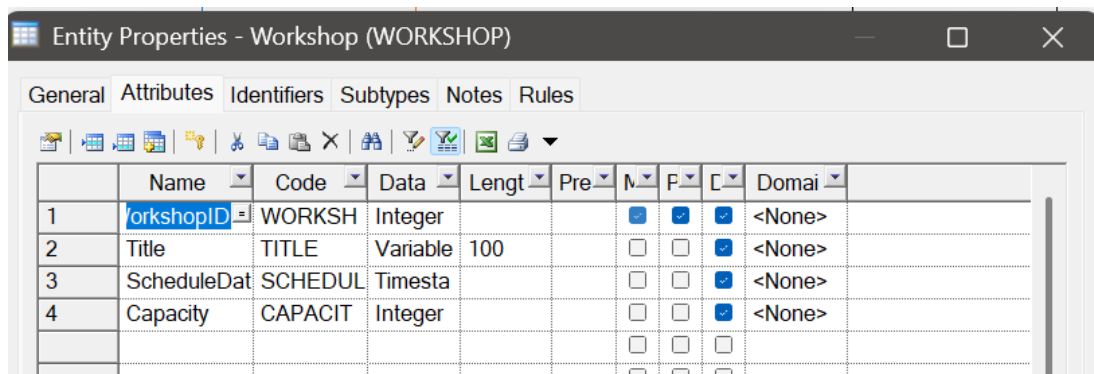


Entity Properties - Studio (STUDIO)

General Attributes Identifiers Subtypes Notes Rules

	Name	Code	P
1	StudioID	STUDIOID	☒
			☐
			☐
			☐

3. Workshop:



Entity Properties - Workshop (WORKSHOP)

General Attributes Identifiers Subtypes Notes Rules

	Name	Code	Data	Length	Pre	M	P	C	Domain
1	WorkshopID	WORKSH	Integer			☒	☒	☒	<None>
2	Title	TITLE	Variable	100		☐	☐	☒	<None>
3	ScheduleDat	SCHEDUL	Timesta			☐	☐	☒	<None>
4	Capacity	CAPACIT	Integer			☐	☐	☒	<None>
						☐	☐	☐	
						☐	☐	☐	

Entity Properties - Workshop (WORKSHOP)				
General Attributes Identifiers Subtypes Notes Rules				
	Name	Code	P	
1	WorkshopID	WORKSHOPID	☒	
			☐	

4.RawMaterial:

Entity Properties - RawMaterial (RAWMATERIAL)										
General Attributes Identifiers Subtypes Notes Rules										
	Name	Code	Data	Length	Pre	M	P	C	Domain	
1	MaterialID	MATERIA	Integer			☒	☒	☒	<None>	
2	RawMaterial	RAWMAT	Variable	50		☐	☐	☒	<None>	
3	UnitOfMeas	UNITOFM	Variable	20		☐	☐	☒	<None>	
4	StockQuantit	STOCKQU	Decimal	10	2	☐	☐	☒	<None>	
						☐	☐	☐		
						☐	☐	☐		

Entity Properties - RawMaterial (RAWMATERIAL)

GeneralAttributesIdentifiersSubtypesNotesRules



	Name	Code	P	
1	MaterialID	MATERIALID	☒	
			☐	
			☐	

5.Tool:

Entity Properties - Tool (TOOL)										
General Attributes Identifiers Subtypes Notes Rules										
	Name	Code	Data	Length	Pre	M	P	C	Domain	
1	ToolID	TOOLID	Integer			☒	☒	☒	<None>	
2	Description	DESCRIP	Variable	255		☐	☐	☒	<None>	
3	Condition	CONDITIO	Variable	30		☐	☐	☒	<None>	
4	IsHighEnd	ISHIGHE	Boolean			☐	☐	☒	<None>	
						☐	☐	☐		

Entity Properties - Tool (TOOL)				
General Attributes Identifiers Subtypes Notes Rules				
	Name	Code	P	
1	ToolID	TOOLID	☒	
			☐	
			☐	

6. ResidentArtist:

Entity Properties - ResidentArtist (RESIDENTARTIST)										
General Attributes Identifiers Subtypes Notes Rules										
	Name	Code	Data	Length	Pre	N	P	C	Domain	
1	ArtistID	ARTISTID	Integer			☒	☒	☒	<None>	
2	ResidentNa	RESIDEN	Variable	100		☐	☐	☒	<None>	
3	Specialty	SPECIALT	Variable	50		☐	☐	☒	<None>	
4	ResidentEm	RESIDEN	Variable	255		☐	☐	☒	<None>	
						☐	☐	☐		

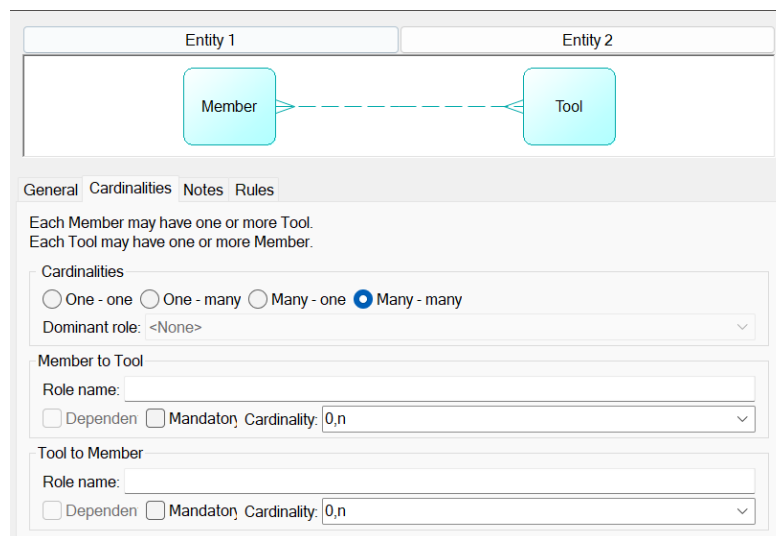
Entity Properties - ResidentArtist (RESIDENTARTIST)				
General Attributes Identifiers Subtypes Notes Rules				
	Name	Code	P	
1	ArtistID	ARTISTID	☒	
			☐	

1-Relationships

[illegible]

Borrows: Member to Tool

- **Type: Many-to-Many (M:N)**



- **Meaning: One Member can borrow and use many different tools, and one Tool can be used by many different members over its lifespan.**

Consumes : Workshop to RawMaterial

- **Type: Many-to-Many (M:N)**

The diagram shows two entities, Workshop and RawMaterial, connected by a dashed line with crow's foot notation. Workshop is on the left and RawMaterial is on the right. Both ends of the line have a crow's foot symbol, indicating a mandatory many-to-many relationship.

General Cardinalities Notes Rules

Each Workshop may have one or more RawMaterial.
Each RawMaterial may have one or more Workshop.

Cardinalities

☐ One - one ☐ One - many ☐ Many - one ☒ Many - many

Dominant role: <None>

Workshop to RawMaterial

Role name:

☐ Dependent ☐ Mandatory Cardinality: 0,n

RawMaterial to Workshop

Role name:

☐ Dependent ☐ Mandatory Cardinality: 0,n

- **Meaning: One Workshop uses many kinds of raw materials, and one Raw Material can be used across many different workshops.**

Hosts : Studio to Workshop

- **Type: One-to-Many (1:N)**

The diagram shows two entities, Studio and Workshop, connected by a dashed line. Studio is on the left and Workshop is on the right. The line has an arrowhead pointing towards Workshop, indicating a one-to-many relationship.

Entity 1: Studio
Entity 2: Workshop

General Cardinalities Notes Rules

Each Studio may have one or more Workshop.
Each Workshop must have one and only one Studio.

Cardinalities
☐ One - one ☒ One - many ☐ Many - one ☐ Many - many
Dominant role: <None>

Studio to Workshop
Role name:
☐ Dependence ☐ Mandatory Cardinality: 0,n

Workshop to Studio
Role name:
☐ Dependence ☒ Mandatory Cardinality: 1,1

- **Meaning: One Studio can host many different workshops over time, but each Workshop must take place in exactly one dedicated studio.**

Instructs : ResidentArtist to Workshop

- **Type: One-to-Many (1:N)**

The screenshot displays a software interface for defining a relationship between two entities, Entity 1 and Entity 2. Entity 1 is labeled 'ResidentArtist' and Entity 2 is labeled 'Workshop'. A dashed line connects them, with an arrow pointing towards 'Workshop'.

Below the diagram, there are tabs for 'General', 'Cardinalities', 'Notes', and 'Rules'. The 'General' tab is selected, showing the following text:

Each ResidentArtist may have one or more Workshop.
Each Workshop must have one and only one ResidentArtist.

Under the 'Cardinalities' section, there are four radio buttons: 'One - one', 'One - many' (selected), 'Many - one', and 'Many - many'. Below this is a 'Dominant role' dropdown menu set to '<None>'. There are two sections for role definitions:

ResidentArtist to Workshop
Role name: _____
☐ Dependence ☐ Mandatory Cardinality: 0,n

Workshop to ResidentArtist
Role name: _____
☐ Dependence ☒ Mandatory Cardinality: 1,1

- **Meaning: One Resident Artist can lead many different workshops, but each Workshop must be led by exactly one artist.**

Registers : Member to Workshop

- **Type: Many-to-Many (M:N)**

The image shows a software configuration interface for a Many-to-Many relationship between two entities, 'Member' and 'Workshop'.

Diagram: At the top, a diagram shows 'Entity 1' containing a 'Member' entity and 'Entity 2' containing a 'Workshop' entity. A dashed line with crow's foot notation connects them, indicating a many-to-many relationship.

Configuration Panel: Below the diagram is a configuration panel with tabs for 'General', 'Cardinalities', 'Notes', and 'Rules'. The 'General' tab is active.

- Text:** 'Each Member may have one or more Workshop.' and 'Each Workshop may have one or more Member.'
- Cardinalities:** Radio buttons for 'One - one', 'One - many', 'Many - one', and 'Many - many'. The 'Many - many' option is selected.
- Dominant role:** A dropdown menu currently showing '<None>'.
- Member to Workshop:** A section with a 'Role name:' field, and checkboxes for 'Dependent' and 'Mandatory'. The 'Cardinality' is set to '0,n'.
- Workshop to Member:** A section with a 'Role name:' field, and checkboxes for 'Dependent' and 'Mandatory'. The 'Cardinality' is set to '0,n'.

- **Meaning: One Member can register for and attend many workshops, and one Workshop can be attended by many different members.**

2.Functional Requirements

- . Member Management: The system must store member contact details and track their subscription payment status.**
- . Workshop Scheduling: The system must allow users to create workshops with a specific title, date, and maximum capacity.**
- . Resource Tracking: The system must maintain an inventory of tools (including their condition) and raw materials (including stock quantities).**
- . Registration System: The system must allow members to sign up for workshops and record which tools they borrow.**

3.Business Rules

- **Studio Constraint: A workshop cannot exist without a physical studio assigned to it, and a studio can host multiple workshops over time.**
- **Instruction Rule: Every workshop must be led by exactly one resident artist.**
- **Participation Rule: One member can attend many workshops, and one workshop can have many different members attending**
- **Tool Usage: A member can borrow multiple tools, and a single tool can be used by many different members over its lifetime**
- **Material Usage: Each workshop requires multiple types of raw materials, and the same material type can be used in different workshops**